COMPUTER FUNDAMENTALS · STUDY GUIDE

Introduction to Virus and Vaccines, Applications, DTP, E-Mail and Internet

A complete, exam-focused reference covering computer security, software categories, desktop publishing, electronic mail, and the internet.

- Computer Viruses
- Antivirus / Vaccines
- Application Software
- Desktop Publishing (DTP)
- Electronic Mail (E-Mail)
- The Internet

STUDY NOTES | DIAGRAMS | KEY TERMS | REVISION QUESTIONS

CHAPTER 1

Computer Virus

1.1 What is a Computer Virus?

> **DEFINITION**
>
> A **computer virus** is a malicious program or piece of code that is loaded onto a computer without the user's knowledge and runs against the user's wishes. It can replicate itself by attaching its copies to other programs or files, and it spreads from one computer to another in a manner similar to a biological virus.

A virus needs a **host program** to attach itself to — it cannot run independently. Once executed, it can corrupt data, slow down the system, steal information, or damage files and the operating system itself.

Key Characteristics of a Virus

- **Self-replicating** — makes copies of itself to spread further.
- **Requires a host** — attaches to executable files, documents, or boot sectors.
- **Triggered by an event** — activates when a date arrives, a file is opened, or a condition is met (the "trigger").
- **Harmful payload** — performs damaging actions such as deleting files or corrupting data.
- **Stealthy** — often tries to hide from the user and from antivirus scans.

1.2 History of Computer Viruses

The concept of a self-replicating program was first theorized academically in the early 1980s. The term "computer virus" was coined by **Fred Cohen** in 1983 while demonstrating such a program as part of his research. Early viruses such as **"Elk Cloner"** (1982, Apple II) and **"Brain"** (1986, the first IBM PC virus) spread through floppy disks. As computers became networked and later connected to the internet, viruses evolved from spreading via removable media to spreading through email attachments, downloads, and network vulnerabilities.

1.3 How a Virus Works (Lifecycle)

A typical virus passes through four stages during its life on an infected system:

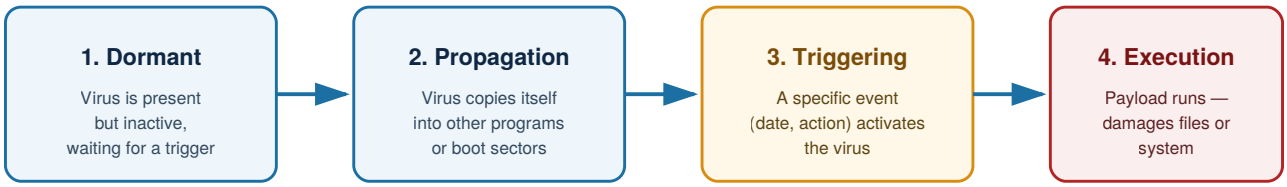


Fig 1.1 — The four-stage lifecycle of a computer virus

1.4 Types of Computer Viruses

Boot Sector Virus
Infects the master boot record of a disk; activates when the system is started from the infected disk.

File / Program Virus
Attaches itself to executable files (.exe, .com) and activates when the infected program is run.

Macro Virus
Written in a macro language and embedded in documents (e.g., Word, Excel); runs when the document is opened.

Polymorphic Virus
Changes its own code/signature each time it replicates, making it harder for antivirus software to detect.

Resident Virus
Installs itself in the computer's memory (RAM) and can infect files even after the original program has closed.

Multipartite Virus
Infects both the boot sector and program files simultaneously, spreading in multiple ways at once.

1.5 Symptoms of Virus Infection

- Computer runs unusually **slow** or freezes/crashes frequently.
- Programs **open or close on their own**, or fail to start at all.
- Unexpected **pop-ups**, error messages, or strange screen behaviour.
- Files go **missing, get renamed, or become corrupted** without reason.
- Unusual increase in **network/internet activity** or data usage.
- **Antivirus software gets disabled** automatically or cannot be updated.
- Storage space **decreases rapidly** for no clear reason.

1.6 Sources of Virus Infection

Source	How Infection Happens
USB / Removable Drives	Plugging in an infected pen drive or external disk
Email Attachments	Opening infected files attached to emails
Internet Downloads	Downloading software/files from untrusted websites
Network Connections	Spreading through shared/unsecured networks
Pirated Software	Installing cracked or unlicensed software

KEY TERMS

Virus — self-replicating malicious code that needs a host file.

Payload — the actual damaging action a virus performs.

Trigger — the event/condition that activates the virus.

Host file — the program/file a virus attaches itself to.

QUICK REVISION POINTS

- A virus is self-replicating, needs a host, and has a trigger + payload.
- Lifecycle: Dormant → Propagation → Triggering → Execution.
- Main types: Boot sector, File, Macro, Polymorphic, Resident, Multipartite.
- Common entry points: USB drives, email attachments, downloads, networks.

CHAPTER 2

Vaccines / Antivirus Software

2.1 What is Antivirus Software?

DEFINITION

Antivirus software (sometimes called a computer "vaccine") is a program designed to detect, prevent, and remove malicious software including viruses, worms, trojans, and other malware from a computer system. Just as a biological vaccine helps the body recognise and fight a disease, antivirus software helps a computer recognise and neutralise harmful code before it causes damage.

Antivirus software runs continuously in the background (real-time protection) and can also perform full system scans on demand or on a schedule.

2.2 Detection Methods

Signature-Based Detection

Compares files against a database of known virus "signatures" (unique code patterns). Fast and accurate for known viruses, but needs regular updates.

Heuristic Detection

Examines the behaviour and structure of a program to identify suspicious activity, allowing detection of new or modified viruses not yet in the database.

Sandboxing

Runs suspicious programs in an isolated virtual environment to observe their behaviour safely before allowing them on the real system.

2.3 How Antivirus Works (Process Flow)

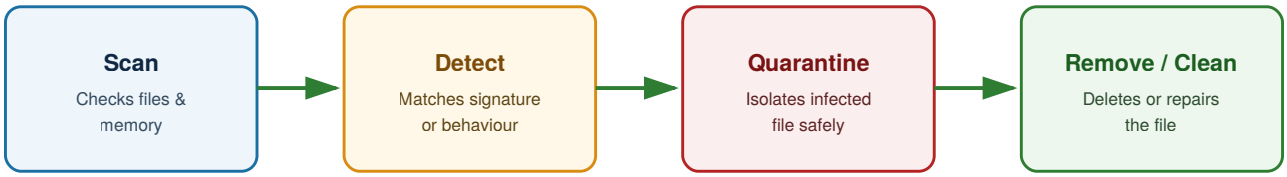


Fig 2.1 — Typical antivirus software working process

- **Scan** — checks files and memory for threats.
- **Detect** — matches a signature or spots suspicious behaviour.
- **Quarantine** — isolates the infected file so it can't spread.
- **Remove / Clean** — deletes or repairs the infected file.

2.4 Popular Antivirus Software

Antivirus	Notable Feature
Windows Defender (Microsoft Defender)	Built-in to Windows, free, real-time protection
Avast / AVG	Free and premium versions, widely used
Norton / McAfee	Comprehensive security suites with firewall, VPN add-ons
Kaspersky	Strong malware detection rates
Quick Heal	Popular in India, lightweight scanning

2.5 Best Practices for Prevention

DO

- Keep antivirus software updated
- Run regular full-system scans
- Update OS and applications regularly
- Scan USB drives before opening
- Back up important data regularly

DON'T

- Open email attachments from unknown senders
- Download software from untrusted websites
- Use pirated/cracked software
- Disable antivirus or firewall protection
- Click on suspicious links or pop-ups

KEY TERMS

Signature — a unique identifying pattern of known malware code.

Heuristic analysis — detecting malware by behaviour rather than known signatures.

Quarantine — isolating a suspicious/infected file so it cannot cause harm.

Real-time protection — continuous background monitoring for threats.

QUICK REVISION POINTS

- Antivirus = "vaccine" for computers; detects, prevents, removes malware.
- Two main detection methods: signature-based and heuristic.
- Process flow: Scan → Detect → Quarantine → Remove/Clean.
- Prevention is as important as detection — safe habits reduce risk greatly.

CHAPTER 3

Applications (Application Software)

3.1 What is Application Software?

DEFINITION

Application software refers to programs designed to help users perform specific tasks — such as writing documents, managing finances, editing images, browsing the internet, or communicating with others. Unlike the operating system, which runs in the background to manage the computer, application software is what the user directly interacts with to get real work done.

3.2 Categories of Application Software

Word Processing Software

Used to create, edit, and format text documents. Examples: MS Word, Google Docs, LibreOffice Writer.

Spreadsheet Software

Used for numerical data, calculations, and charts in rows/columns. Examples: MS Excel, Google Sheets.

Presentation Software

Used to create slide-based visual presentations. Examples: MS PowerPoint, Google Slides.

Database Software

Used to store, organise, and retrieve large amounts of structured data. Examples: MS Access, MySQL, Oracle.

Multimedia Software

Used to create or play audio, video, animation, and graphics. Examples: VLC Media Player, Adobe Premiere, Photoshop.

Web Browsers

Used to access and view content on the World Wide Web. Examples: Chrome, Firefox, Edge.

3.3 Comparison Table: System vs Application Software

Basis	System Software	Application Software
Purpose	Manages and operates hardware/ system resources	Helps users perform specific tasks
Dependency	Required for the computer to function at all	Runs on top of system software; optional
Examples	Windows, Linux, macOS, device drivers	MS Word, Excel, PowerPoint, browsers

Basis	System Software	Application Software
Installation	Pre-installed, essential to the device	Installed separately as needed by the user
User interaction	Mostly runs in the background	Directly interacted with by the user

KEY TERMS

Application software — software for performing user-specific tasks.

System software — software that manages computer hardware/resources.

GUI — Graphical User Interface used by most application software.

QUICK REVISION POINTS

- Application software performs specific user tasks; system software runs the machine.
- Main categories: word processing, spreadsheet, presentation, database, multimedia, browsers.
- Application software depends on system software to function.

CHAPTER 4

DTP (Desktop Publishing)

4.1 What is DTP?

> **DEFINITION**
>
> **Desktop Publishing (DTP)** means creating neat, professional-looking documents using a computer and special design software — like newsletters, brochures, magazines, and books — by combining text and pictures together on the page.

DTP software gives precise control over page layout, typography, colour, and image placement that general word processors do not offer to the same degree.

4.2 Features of DTP Software

- **Page layout tools** — precise control over margins, columns, and grids.
- **Typography controls** — fine adjustment of fonts, kerning, and line spacing.
- **Image & text frame handling** — place, resize, and wrap text around images easily.
- **Templates** — pre-built layouts for common publication types.
- **Colour management** — accurate colour control for print output (CMYK support).
- **Multi-page document handling** — master pages, page numbering, table of contents.

4.3 Popular DTP Software

Software	Notes
Adobe InDesign	Industry-standard professional DTP tool
Adobe PageMaker	Earlier widely-used DTP software (now discontinued)
MS Publisher	Beginner-friendly DTP tool bundled with MS Office
CorelDraw	Vector graphics and layout design tool
Scribus	Free, open-source DTP software

4.4 Applications of DTP

Newsletters
Regular organisational or club bulletins

Brochures & Flyers
Marketing and promotional material

Books & Magazines
Long-form multi-page publications

Posters & Banners
Large-format visual displays

Business Cards
Small precision-layout print items

Forms & Certificates
Structured official documents

4.5 DTP Page Layout Structure

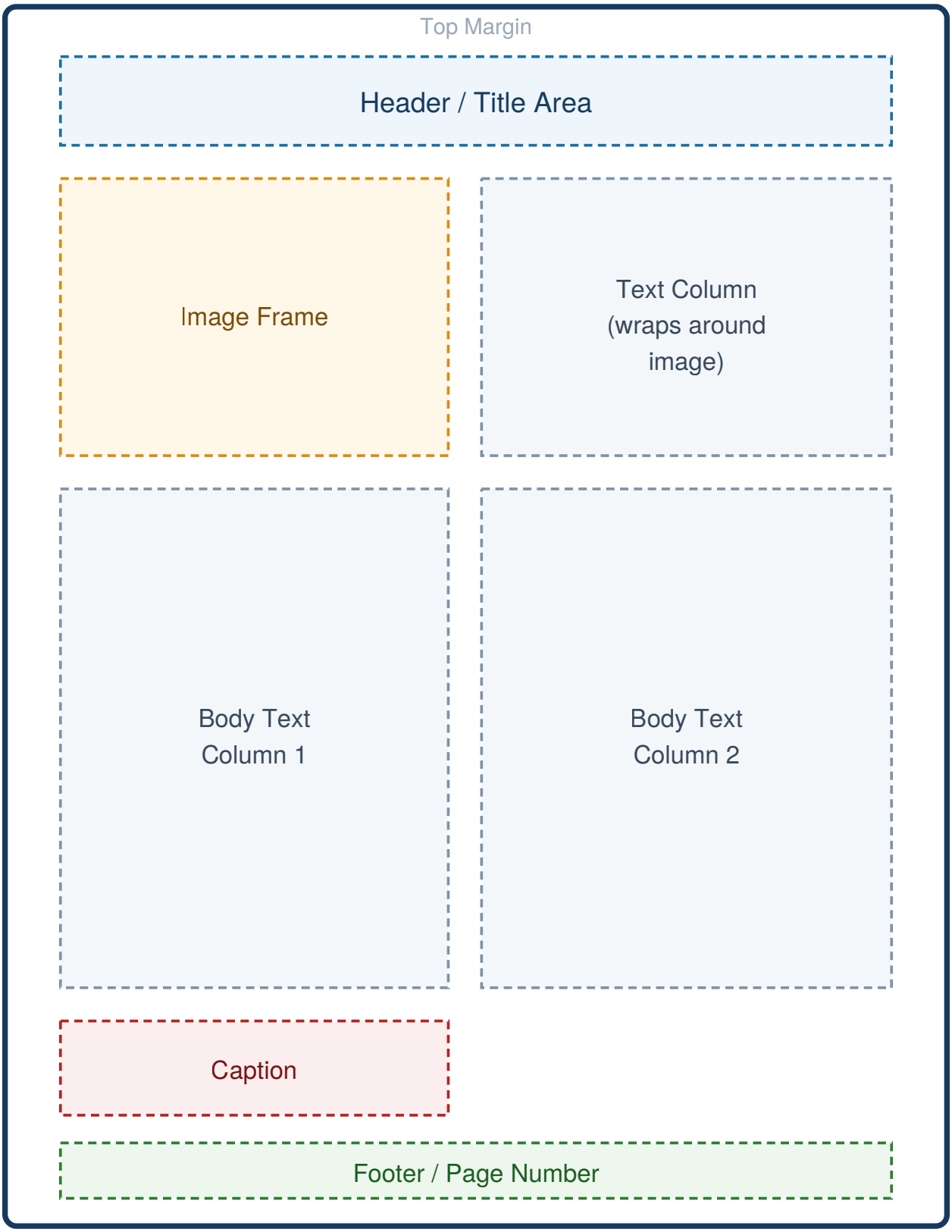


Fig 4.1 — Typical DTP page layout: header, image frame, text columns, caption, and footer regions

KEY TERMS

DTP — Desktop Publishing; computer-based page design and layout.

Master page — a template layout reused across multiple pages of a document.

CMYK — the colour model (Cyan, Magenta, Yellow, Key/Black) used for print.

QUICK REVISION POINTS

- DTP = combining text and graphics digitally for professional print/digital output.
- Key features: page layout tools, typography control, templates, colour management.
- Common tools: Adobe InDesign, MS Publisher, CorelDraw, Scribus.
- Used for newsletters, brochures, books, posters, and more.

CHAPTER 5

E-Mail

5.1 What is E-Mail?

DEFINITION

Electronic Mail (E-Mail) is a method of exchanging digital messages between people using electronic devices over a network such as the internet. It allows text, images, documents, and other files to be sent almost instantly to one or more recipients anywhere in the world.

5.2 How E-Mail Works (Architecture)

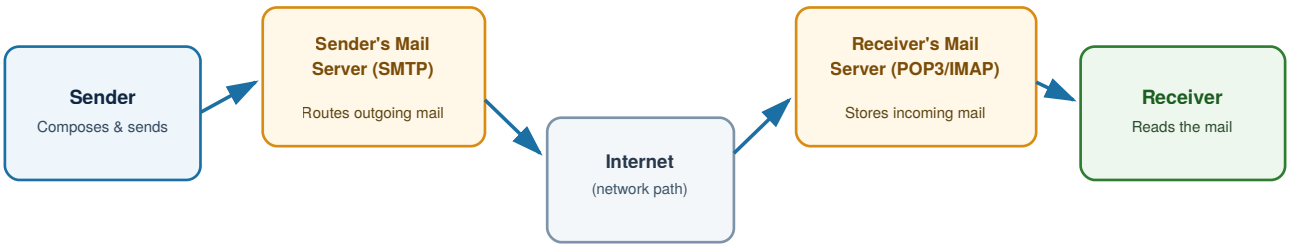


Fig 5.1 — Path of an email from sender to receiver through mail servers

The sender's mail client connects to an outgoing mail server (using the **SMTP** protocol) which routes the message across the internet to the receiver's mail server. The message is stored there until the receiver's mail client retrieves it using **POP3** or **IMAP** protocols.

5.3 Structure of an E-Mail

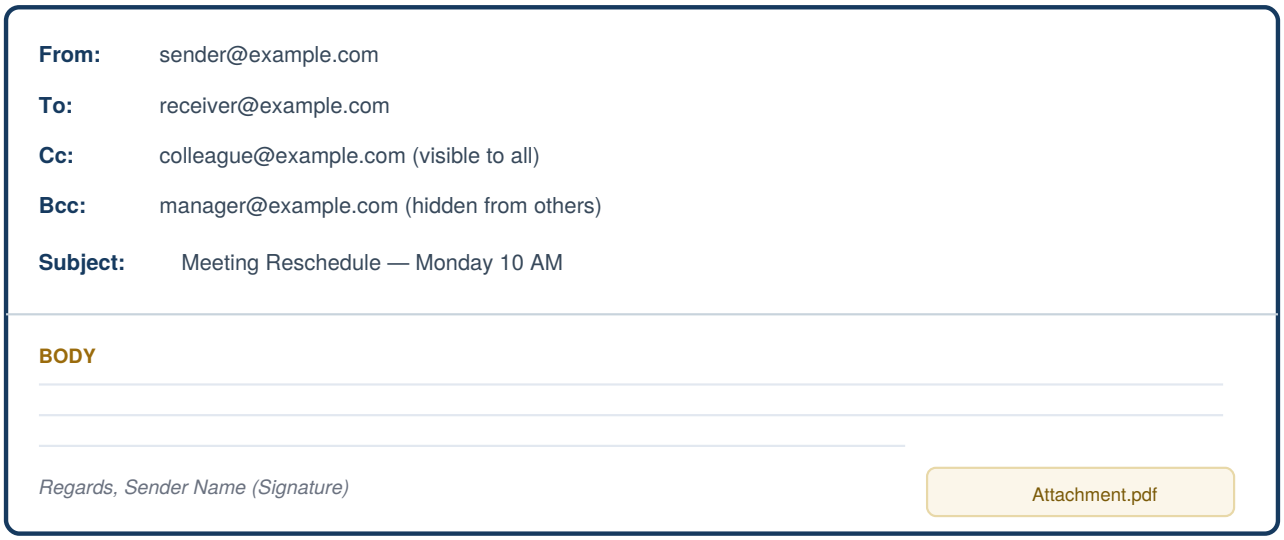


Fig 5.2 — Labelled structure of a typical email

Field	Purpose
To	Primary recipient(s) of the email
Cc (Carbon Copy)	Additional recipients; visible to everyone
Bcc (Blind Carbon Copy)	Additional recipients; hidden from other recipients
Subject	Short summary of the email's content
Body	The main message content
Attachment	Files (documents, images) sent along with the email
Signature	Sender's name/contact details at the end of the message

5.4 Advantages and Disadvantages of E-Mail

ADVANTAGES

- Instant, fast delivery worldwide
- Low cost compared to postal mail
- Can send documents, images, and files
- Easy to keep records/search old messages
- Eco-friendly — reduces paper use

DISADVANTAGES

- Risk of viruses via attachments
- Spam and phishing emails
- Requires internet access and a device
- Possible miscommunication (no tone/body language)
- Information overload from large inboxes

5.5 E-Mail Etiquette

- Always use a clear and relevant **subject line**.
- Use a polite greeting and closing/signature.
- Avoid using ALL CAPS (can appear as shouting).
- Proofread before sending; check attachments are included.
- Use Cc/Bcc appropriately and avoid unnecessary "reply all".

KEY TERMS

SMTP — Simple Mail Transfer Protocol, used to send email.

POP3 / IMAP — protocols used to retrieve/download received email.

Cc / Bcc — carbon copy / blind carbon copy recipient fields.

Spam — unwanted/unsolicited bulk email.

QUICK REVISION POINTS

- Email travels: Sender → Sender's mail server (SMTP) → Internet → Receiver's mail server → Receiver.
- Key fields: To, Cc, Bcc, Subject, Body, Attachment, Signature.
- Advantages: speed, low cost, file sharing. Disadvantages: spam, virus risk.

CHAPTER 6

Internet

6.1 What is the Internet?

> **DEFINITION**
>
> The **Internet** is a vast global network of interconnected computers and servers that communicate using a standard set of protocols (**TCP/IP**), allowing the exchange of information and services such as web browsing, email, and file transfer across the world.

6.2 Internet vs Intranet vs WWW

Term	Meaning
Internet	Global public network connecting millions of devices worldwide
Intranet	Private network restricted to an organisation, using internet-like technology
WWW (World Wide Web)	A system of interlinked web pages/documents accessed via the internet using browsers

The internet is the underlying network infrastructure, while the WWW is one of the many services that run on top of it.

6.3 Brief History of the Internet

The internet originated from **ARPANET**, a US Department of Defense project started in 1969 to connect research computers. The introduction of **TCP/IP** protocols in the 1980s allowed different networks to interconnect. In 1991, **Tim Berners-Lee** introduced the World Wide Web, making the internet accessible and easy to use through web pages and hyperlinks, leading to its rapid global growth through the 1990s and beyond.

6.4 How the Internet Works

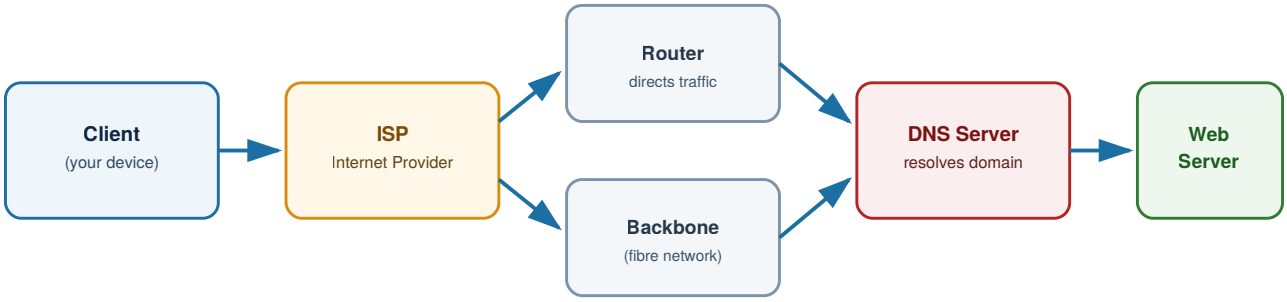


Fig 6.1 — How a request travels from a client device to a web server and back (client-server model)

When you request a website, your device (client) sends the request through your **ISP (Internet Service Provider)**, which routes it across routers and backbone networks. A **DNS server** converts the human-readable domain name into an IP address, and the request finally reaches the appropriate **web server**, which sends the requested data back along the same path.

6.5 Internet Services

World Wide Web (WWW)
Browsing interlinked web pages and websites

E-Mail
Sending and receiving digital messages

FTP
Transferring files between computers over a network

Chat / Messaging
Real-time text communication (WhatsApp, Telegram)

Social Media
Platforms for sharing content and networking

Video Conferencing
Real-time audio-video communication (Zoom, Teams)

Cloud Storage
Storing and accessing files online (Google Drive, Dropbox)

E-Commerce
Buying and selling goods/services online

6.6 Advantages, Risks and Safety Tips

ADVANTAGES

- Instant access to global information
- Enables communication and collaboration
- Supports online education and remote work
- E-commerce and online banking convenience

RISKS

- Exposure to viruses and malware
- Phishing, scams, and identity theft
- Privacy concerns and data breaches
- Cyberbullying and inappropriate content

INTERNET SAFETY TIPS

Use strong, unique passwords; keep antivirus software updated (see Chapter 2); avoid clicking unknown links or attachments; verify website security (HTTPS) before entering personal data; and be cautious about sharing personal information on social media.

KEY TERMS

TCP/IP — the core protocol suite that governs data transmission on the internet.

ISP — Internet Service Provider; gives users access to the internet.

DNS — Domain Name System; converts domain names into IP addresses.

URL — Uniform Resource Locator; the address of a web resource.

QUICK REVISION POINTS

- Internet = global network; WWW = a service (web pages) that runs on it; Intranet = private internal network.
- Originated from ARPANET (1969); WWW introduced by Tim Berners-Lee (1991).
- Working: Client → ISP → Router/Backbone → DNS → Web Server.
- Key services: WWW, Email, FTP, Chat, Social Media, Video Conferencing, Cloud Storage.
- Stay safe: updated antivirus, strong passwords, avoid suspicious links.